2(a) The box is initially at rest. The tension in the spring initially balances the total weight of the box and the plasticine. When the plasticine falls, <u>the tension is greater than the weight of the box</u>. B1 There is a <u>restoring force/ resultant force upwards, towards an equilibrium position</u>. B1

When the extension of the spring is reduced, the weight is greater than the tension in the spring. There is a <u>restoring force/ resultant force downwards, towards an equilibrium position.</u> B1

Hence, the box oscillates about an equilibrium position.

2(b)(i) F = (0.02)(9.81)

$$a = \frac{F}{m} = \frac{(0.02)(9.81)}{0.100}$$ M1

$$= 1.96 \text{ m s}^{-2}$$ A1

2(b)(ii) $|a| = \omega^2 x_o$

$x_o = 2.50$ cm

$$\omega^2 = \frac{a}{x_o} = \frac{1.962}{2.5 \times 10^{-2}}$$ M1

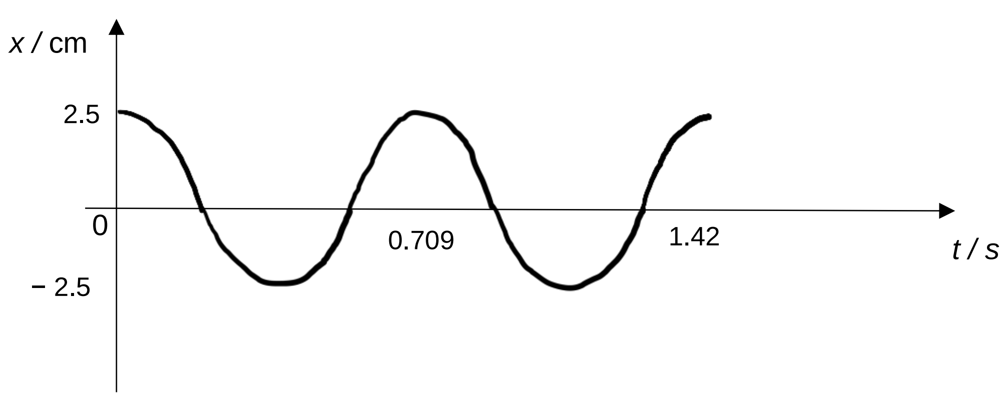
$$\omega = \sqrt{\frac{1.962}{2.5 \times 10^{-2}}} = 8.8589 \text{ rad s}^{-1}$$

$$T = \frac{2\pi}{\omega}$$ M1

$$= \frac{2\pi}{8.8589}$$

$$= 0.709 \text{ s}$$ A1

2(c) B1 cosine variation



B1 label correct period and amplitude values

DO NOT WRITE IN THIS MARGIN

3 **(a)** Electric potential at a point is defined as the work done per unit positive charge by B1